



SFM-01 Introduction to Excel

Overview

In this session, the faculty will guide you through various strategies using mathematical formulae and Excel functions, starting with short-term trading strategies such as carrying overnight positions and buying at today's close and selling at tomorrow's open. He will explain the concepts of moving averages, mean reversion, Volume Weighted Average Price (VWAP), and the Average True Range (ATR).

Pay close attention to this hands-on session since your assignments will be based on the concepts covered.

Key takeaways:

After completing this module, you will be able to:

- Use Excel for quantitative calculations.
- Systematically formulate trading strategies on Excel.
- Perform mathematical functions such as sum, sum-if, lookup/search function (such as offset, index, match), and formatting.

Pre-requisites:

- Basic knowledge of MS Excel or knowledge of any other financial computing tool is preferable.
- Completion of primer tests 2 (Stock Market Basics) and 3 (Excel) is mandatory.
- Ability to work with the raw data in the file given: "XLP.csv".
- Perform the following tasks on the same sheet:
 - a. Freeze the top row of column names
 - b. Find the difference in current day prices over the previous day prices
 - c. If prices are increasing, add label "Increase", else "Decrease"

Practical use of the topics learnt in this session

Post this lecture, you will be in a position to:

- Work on essential mathematical Excel functions like sum, sumif, which will help you in coding strategy backtests, eventually
- Use Excel to calculate technical indicators
- Learn different strategies and quantify them using Excel

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- Learn about Simple Moving Average to normalize price movements and for better market trend analysis
- Calculate trade returns, positive and negative trades and hit ratio to find out profitability
- Calculate the fair price of the asset using the Volume Weighted Average Price (VWAP)
- Formulate a breakout strategy on Excel using the Average True Range (ATR)

These concepts are utilized to develop a framework that combines market knowledge, mathematical calculations, and programming, using Excel.

Downloadable files:

- One folder is available for download: “SFM-01-Inclass Exercises File.zip” Please download it prior to the lecture.

Additional reading material:

- [Backtesting Long Short Moving Average Crossover Strategy in Excel](#)
- [Momentum-Based Strategies for Low and High-Frequency Trading – \[EXCEL MODEL\]](#)

Recommended time for the session & related coursework - 8 hours